

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Area and volume

02

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**C**C. $2\sqrt{\pi}$

Choice C is correct. The area A of a circle with radius r is given by the formula $A = \pi r^2$. Thus, a circle with radius 2 has area $\pi(2^2)$, which can be rewritten as 4π . The area of a square with side length s is given by the formula $A = s^2$. Thus, if a square has the same area as a circle with radius 2, then $s^2 = 4\pi$. Since the side length of a square must be a positive number, taking the square root of both sides of $s^2 = 4\pi$ gives $s = \sqrt{4\pi}$. Using the properties of square roots, $\sqrt{4\pi}$ can be rewritten as $(\sqrt{4})(\sqrt{\pi})$, which is equivalent to $2\sqrt{\pi}$. Therefore, $s = 2\sqrt{\pi}$.

Choice A is incorrect. The side length of the square isn't equal to the radius of the circle.

Choices B and D are incorrect and may result from incorrectly simplifying the expression $\sqrt{4\pi}$.

Q2**4.41**

Accept also: 441/100

The correct answer is 4.41. The area, A , of a circle is given by the formula $A = \pi r^2$, where r is the radius of the circle. It's given that the area of the circle is $b\pi$ square inches, where b is a constant, and the radius of the circle is 2.1 inches. Substituting $b\pi$ for A and 2.1 for r in the formula $A = \pi r^2$ yields $b\pi = \pi(2.1)^2$. Dividing both sides of this equation by π yields $b = 4.41$. Therefore, the value of b is 4.41.

Q3**A****A.** 113π

Choice A is correct. It's given that circle N has a radius of 7 mm. The area of a circle is given by the expression πr^2 , where r is the radius of the circle. Substituting 7 for r in the expression yields $\pi(7)^2$, or 49π , for the area, in mm^2 , of circle N. It's also given that the area of circle M is $64\pi \text{ mm}^2$. Adding the two areas yields $49\pi + 64\pi$, or 113π , mm^2 . Therefore, the total area, in mm^2 , of circles N and M is 113π .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**D****D.** 116π

Choice D is correct. The area, A , of a circle is given by the formula $A = \pi r^2$, where r represents the radius of the circle. It's given that circle K has a radius of 4 millimeters mm. Substituting 4 for r in the formula $A = \pi r^2$ yields $A = \pi 4^2$, or $A = 16\pi$. Therefore, the area of circle K is $16\pi \text{ mm}^2$. It's given that circle L has an area of $100\pi \text{ mm}^2$. Therefore, the total area, in mm^2 , of circles K and L is $16\pi + 100\pi$, or 116π .

Choice A is incorrect. This is the sum of the radii, in mm, of circles K and L multiplied by π , not the total area, in mm^2 , of the circles.

Choice B is incorrect. This is the sum of the diameters, in mm, of circles K and L multiplied by π , not the total area, in mm^2 , of the circles.

Choice C is incorrect and may result from conceptual or calculation errors.

Q5**B****B.** 350

Choice B is correct. The area, A , of a triangle is given by $A = \frac{1}{2}bh$, where b is the length of a base of the triangle and h is the corresponding height of the triangle. It's given that a triangle has a base length of 10 centimeters and a corresponding height of 70 centimeters. Substituting 10 for b and 70 for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}1070$, or $A = 350$. Therefore, the area, in square centimeters, of the triangle is 350.

Choice A is incorrect. This is the product of the given base and height of the triangle, not its area.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the sum of the given base and height of the triangle, not its area.

Q6**D****D.** $1,152\pi$

Choice D is correct. The volume, V , of a right circular cylinder is given by $V = \pi r^2 h$, where r is the radius of the circular base and h is the height of the cylinder. It's given that the cylinder has a height of 8 meters and a base with a radius of 12 meters. Substituting 12 for r and 8 for h in $V = \pi r^2 h$ yields $V = \pi(12)^2 8$, or $V = 1,152\pi$. Therefore, the volume, in m^3 , of the cylinder is $1,152\pi$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the volume, in m^3 , of a cylinder with a radius of 8 meters and a height of 12 meters.

Q7**C****C. 640 cm^3**

Choice C is correct. The total volume of the cylindrical can is found by multiplying the area of the base of the can, 75 cm^2 , by the height of the can, 10 cm, which yields 750 cm^3 . If the syrup needed to fill the can has a volume of 110 cm^3 , then the remaining volume for the pieces of fruit is $750 - 110 = 640 \text{ cm}^3$.

Choice A is incorrect because if the fruit had a volume of 7.5 cm^3 , there would be $750 - 7.5 = 742.5 \text{ cm}^3$ of syrup needed to fill the can to the top. Choice B is incorrect because if the fruit had a volume of 185 cm^3 , there would be $750 - 185 = 565 \text{ cm}^3$ of syrup needed to fill the can to the top. Choice D is incorrect because it is the total volume of the can, not just of the pieces of fruit.

Q8**7.5**

Accept also: 15/2

The correct answer is $\frac{15}{2}$. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. In the right triangle shown, the length of the base of the triangle is 5 inches, and the height is 3 inches. It follows that $b = 5$ and $h = 3$. Substituting 5 for b and 3 for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}53$, which is equivalent to $A = \frac{1}{2}15$, or $A = \frac{15}{2}$. Therefore, the area of the triangle, in square inches, is $\frac{15}{2}$. Note that 15/2 and 7.5 are examples of ways to enter a correct answer.

Q9**C****C.** 726π

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of 6 centimeters. Therefore, $h = 6$. It's also given that the cylinder has a base diameter of 22 centimeters. The radius of a circle is half the diameter of the circle. Since the base of a right circular cylinder is a circle, it follows that the radius of the base of the right circular cylinder is $\frac{22}{2}$, or 11, centimeters. Therefore, $r = 11$. Substituting 11 for r and 6 for h in the formula $V = \pi r^2 h$ yields $V = \pi 11^2 6$, which is equivalent to $V = \pi 121 6$, or $V = 726\pi$. Therefore, the volume, in cubic centimeters, of the cylinder is 726π .

Choice A is incorrect. This is the volume of a right circular cylinder that has a base diameter of $2\sqrt{22}$, not 22, centimeters and a height of 6 centimeters.

Choice B is incorrect. This is the volume of a right circular cylinder that has a base diameter of $4\sqrt{11}$, not 22, centimeters and a height of 6 centimeters.

Choice D is incorrect. This is the volume of a right circular cylinder that has a base diameter of 44, not 22, centimeters and a height of 6 centimeters.

Q10**D****D.** $94 \cdot 94$

Choice D is correct. The area of a square is given by s^2 , where s is the length of each side of the square. It's given that the length of each side of a square is 94 cm. It follows that the area, in cm^2 , of the square is 94^2 , or $94 \cdot 94$. Therefore, the expression that gives the area, in cm^2 , of the square is $94 \cdot 94$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This expression gives the perimeter, in cm, of the square.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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